

Shapes.inc D&D Dungeon Master Bot

This project provides a Python-based chatbot that leverages the **Shapes.inc API** to act as a full-featured **Dungeons & Dragons (D&D) Dungeon Master (DM)**. The bot is designed to handle storytelling, manage character sheets, and perform dice rolls through direct messages (DMs).

Features

Feature	Description
Immersive Storytelling	The bot uses a specialized system prompt to maintain a consistent DM persona, describing environments and NPCs vividly.
Dice Rolling Tool	An integrated tool allows the AI to roll dice (e.g., 1d20, 2d6+3) for skill checks, combat, and saving throws.
Character Management	Persistent storage for player character sheets, including stats, HP, inventory, and quest progress.
State Persistence	Game state is saved to a local <code>character_sheets.json</code> file, allowing adventures to continue across sessions.
Tool Integration	Uses the Shapes.inc <code>Tool</code> class to give the AI programmatic access to game mechanics.

Prerequisites

- Python 3.7+
- Shapes.inc API Key:** Obtain your API key from the [Shapes.inc Developer Portal](#).
- Install Dependencies:**

Bash

```
pip install shapesinc -U
```

Setup and Usage

1. Set Environment Variable:

Set your Shapes.inc API key as an environment variable:

```
Bash  
  
export SHAPES_API_KEY='your_api_key_here'
```

2. Run the Bot:

Execute the main script to start the interactive DM session:

```
Bash  
  
python dnd_bot.py
```

3. Interaction:

- The bot will prompt you for a **User ID**. This allows it to load your specific character sheet.
- Once started, you can describe your actions in natural language (e.g., "I search the room for traps" or "I attack the goblin with my longsword").
- The DM will respond, calling the dice roller or updating your character sheet as needed.

Code Structure

- `dnd_bot.py` : The main application logic, including tool definitions, system prompts, and the chat loop.
- `character_sheets.json` : (Generated automatically) Stores player data in a structured format.

Customization

You can customize the DM's personality or the starting adventure by modifying the `DM_SYSTEM_PROMPT` variable in `dnd_bot.py`. This allows you to set different themes, such as high fantasy, gothic horror, or sci-fi RPGs.

References

- [Shapes.inc API Documentation](#) [1]

- [Shapes.inc Developer Portal](#) [2]
 - [D&D 5th Edition Rules](#) [3]
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Author: Manus AI

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